

Testing Pattern-Mediated Indirect Evolutionary Rescue

A predator–prey reaction–diffusion research report

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Contents

1	Introduction	1
1.1	Basic Biological Concepts	1
1.2	Literature Positioning and Research Gap	1
1.3	Core Mathematical Problem	2
2	Predator–Prey Dynamics	3
2.1	The Basic Ecological Problem	3
2.2	Historical and Empirical Motivation	3
2.3	Classical Lotka–Volterra Baseline	4
2.4	Why the Classical Model Is Not Enough	5
3	The Extended Lotka–Volterra Model	5
3.1	From One Prey Type to Two	5
3.2	A Non-Spatial Baseline	6
3.3	Evolutionary Sorting	6
3.4	A Coexistence Caveat	7
3.5	Spatial Extension	8
3.6	Predator Rescue Threshold	8
3.7	Operational Definition of Persistence	9
4	Turing Pattern Formation	9
4.1	Why Diffusion Can Create Patterns	9
4.2	The Two-Variable Calculation	9
4.3	Different Diffusion Rates	10
4.4	Pattern Wavelength and Boundaries	11
4.5	How This Enters the Rescue Problem	11
5	Planned Analysis	12
6	Expected Contributions and Limitations	12

1 Introduction

This process of model building is the art of discarding all but the essentials and focusing on a model simple enough to do the job, but not too hard to see all the way through.

Philip Anderson

1.1 Basic Biological Concepts

We begin by fixing the biological language. A *population* is a group of individuals of the same type or species. In mathematical models, its abundance may be tracked as a function of time, such as $X(t)$, or as a density in space and time, such as $X(x, t)$.

A *prey* population is a population that can be consumed by another population. A *predator* population feeds on prey, and the interaction in which predators consume prey is called *predation*. Predation has opposite immediate effects on the two populations: it decreases prey abundance but can increase predator abundance by providing energy for survival and reproduction.

A *trait* is a biological property of individuals that affects growth, survival, movement, or interactions with other species. In this report, the key prey trait is defence against predation. We distinguish between two prey types:

$$U = \text{undefended prey}, \quad D = \text{defended prey}. \quad (1.1)$$

Undefended prey are easier for predators to consume, but they are assumed to grow or compete better. Defended prey are harder to consume, but defence is costly. This is the defence–growth trade-off that motivates the extended model.

An *eco-evolutionary* model is one in which ecological variables, such as population density, interact with evolutionary variables, such as trait composition. In our setting, the predator may be rescued not because it evolves, but because the prey community shifts toward a more edible composition.

Finally, a *spatial* model tracks where populations are located. This matters because predator persistence depends not only on how much edible prey exists, but also on whether predators and edible prey overlap in space. A *Turing pattern* is a spatial pattern generated by reaction–diffusion dynamics when diffusion destabilizes an otherwise stable homogeneous state [1]. The central question of this report is whether such spatial patterns can change the threshold for predator rescue.

1.2 Literature Positioning and Research Gap

The starting point is the broader problem of *evolutionary rescue*: a population under environmental stress may avoid extinction if evolutionary change raises its mean fitness fast enough [2]. Most evolutionary-rescue theory focuses on the threatened population itself. In multispecies communities, however, a focal species can also be rescued indirectly by evolutionary change in an interacting species. Yamamichi and Miner introduced this mechanism as *indirect evolutionary rescue*: prey evolution can prevent

predator extinction when the prey community shifts toward a more edible composition [3]. Hermann and Becks later tested this idea experimentally in a rotifer–alga system, showing that changes in prey genotype frequency can rescue a predator from extinction [4].

This report builds on that predator–prey setting but focuses on a spatial question. Defended and undefended prey are a natural way to represent the key trait axis because defence typically reduces predation risk but carries a growth or competitive cost. Experiments with rotifer–algal systems show that the form of this trade-off can strongly alter eco-evolutionary dynamics [5], and recent work shows that the range of prey trait variation affects the evolutionary contribution to predator growth rates [6]. These studies motivate treating prey composition as a dynamic variable rather than as a fixed ecological background.

The closest spatial precedent is the reaction–diffusion model of Roy et al., who studied defended and undefended prey, eco-evolutionary feedback, coexistence, persistence, extinction, and Turing pattern formation in a spatial predator–prey system [7]. Their work shows that spatio-temporal patterns can arise from the interaction between ecological and evolutionary components. The present report adds a more targeted threshold question: instead of asking only when patterns occur or which states persist, we ask whether spatial pattern formation changes the maximum predator mortality compatible with indirect rescue. The quantity $\Delta m_c = m_c^{\text{PDE}} - m_c^{\text{ODE}}$ is therefore the operational comparison that distinguishes this report from the broader Roy et al. pattern analysis. Thus “pattern-mediated indirect evolutionary rescue” is used here as a proposed mechanism to test, not as an established subfield.

The classical Lotka–Volterra model provides the baseline language for this question. It is not realistic enough for the final problem, but it clearly displays the predator–prey feedback that all later models retain: prey support predators, while predators suppress prey. We therefore begin with that baseline, identify its limitations, and then extend it to include prey heterogeneity, predator stress, spatial movement, and a rescue-threshold comparison.

1.3 Core Mathematical Problem

The study is not merely about finding Turing patterns. Turing instability is a mechanism that can create spatial structure, but the central mathematical problem is to determine whether such structure changes the predator mortality threshold for indirect evolutionary rescue.

The mathematical object of interest is the difference between the predator persistence threshold in the well-mixed system and the corresponding threshold in the spatial reaction–diffusion system. Let m_c^{ODE} denote the largest predator mortality for which the non-spatial model maintains predator persistence, and let m_c^{PDE} denote the corresponding threshold for the spatial model. The central quantity is

$$\Delta m_c = m_c^{\text{PDE}} - m_c^{\text{ODE}}. \quad (1.2)$$

If $\Delta m_c > 0$, spatial patterning promotes indirect evolutionary rescue. If $\Delta m_c < 0$,

spatial patterning inhibits rescue. If Δm_c changes sign across parameter regimes, then the effect of pattern formation is conditional.

Do Turing-type spatial patterns in defended and undefended prey expand or shrink the mortality range over which predators persist?

2 Predator–Prey Dynamics

2.1 The Basic Ecological Problem

A predator–prey system is the simplest setting in which the growth rate of one population depends directly on the abundance of another. Let $X(t)$ denote prey density and $Y(t)$ predator density. If prey are abundant, predators have more opportunities to feed and reproduce. If predators are abundant, prey suffer more mortality. This creates the feedback loop

more prey $\longrightarrow$ more predator growth $\longrightarrow$ more prey mortality.

After prey decline, predators may also decline because their food supply has been reduced. Predator decline then releases prey from predation pressure. The classical predator–prey problem is to turn this verbal feedback into a dynamical system.

2.2 Historical and Empirical Motivation

The equations are named after Alfred J. Lotka and Vito Volterra, who arrived at the same mathematical structure from different directions. Lotka introduced related equations in the context of chemical reactions and later used them in population dynamics. Volterra developed the predator–prey version after discussions with the marine biologist Umberto D’Ancona, who had studied fish catches in the Adriatic Sea. During the First World War, fishing effort was greatly reduced, yet the proportion of predatory fish in the catch increased. Volterra’s model gave a simple mathematical mechanism for why reduced fishing could increase the average abundance of predators relative to prey [8, 9].

The model is also associated with the famous hare–lynx records of the Hudson’s Bay Company. The recorded numbers of snowshoe hare and lynx pelts show repeated oscillations that resemble predator–prey cycles. These data are not a clean laboratory experiment: pelt records are affected by hunting effort, climate, food availability, disease, and many other variables. Still, they remain a useful empirical picture of the kind of delayed predator–prey oscillation that the model tries to capture. Cleaner cyclic behaviour can be observed in controlled systems, such as laboratory rotifer–algae predator–prey experiments, where environmental conditions and species composition are more tightly constrained [9, 10].

These examples should not be taken as proof that the Lotka–Volterra equations are a complete model of real ecosystems. Rather, they explain why the model became central: it is simple enough to analyze, but rich enough to produce a recognizable ecological story.

2.3 Classical Lotka–Volterra Baseline

The Lotka–Volterra model is the standard first mathematical answer to the predator–prey problem [8, 9, 10]. It is built from four assumptions:

1. The system is well mixed, so predator–prey encounters are proportional to the product XY .
2. In the absence of predators, prey grow exponentially.
3. In the absence of prey, predators decline exponentially.
4. The environment is constant, and there is no trait change, adaptation, age structure, space, or stochastic extinction.

These assumptions are strong, but they give the cleanest possible expression of predator–prey feedback. Combining exponential prey growth, predator mortality, and mass-action predation gives

$$\boxed{\begin{aligned}\frac{dX}{dt} &= \alpha X - \beta XY, \\ \frac{dY}{dt} &= \delta XY - \gamma Y.\end{aligned}} \tag{2.1}$$

Here α is the prey growth rate, β is the attack coefficient on prey, δ converts prey consumption into predator growth, and γ is predator mortality.

The per capita form is

$$\frac{1}{X} \frac{dX}{dt} = \alpha - \beta Y, \quad \frac{1}{Y} \frac{dY}{dt} = \delta X - \gamma. \tag{2.2}$$

This form shows two thresholds. Prey increase when predator density is below α/β , and predators increase when prey density is above γ/δ . Equivalently, the non-trivial nullclines are

$$\dot{X} = 0 \iff Y = \frac{\alpha}{\beta}, \quad \dot{Y} = 0 \iff X = \frac{\gamma}{\delta}. \tag{2.3}$$

They intersect at the coexistence equilibrium

$$(X^*, Y^*) = \left(\frac{\gamma}{\delta}, \frac{\alpha}{\beta} \right), \tag{2.4}$$

besides the extinction equilibrium $(0, 0)$.

The coexistence equilibrium is a center in the classical model, leading to neutral closed cycles in the positive quadrant. This cyclic behaviour is the famous qualitative prediction: prey increase, predators respond, prey are suppressed, predators decline, and prey recover. The result is mathematically elegant, but fragile. Once the model is perturbed, for example by adding density-dependent prey growth, neutral cycling generally disappears [8, 9].

2.4 Why the Classical Model Is Not Enough

The Lotka–Volterra model is valuable because it is transparent. Every term has a clear biological meaning, and the predator–prey feedback is easy to identify. However, it is too simple for the problem we ultimately want to study.

First, prey grow exponentially when predators are absent. This means the model has no resource limitation and no carrying capacity. A natural correction is logistic prey growth, where reproduction slows as prey density approaches an ecological ceiling. This correction is not a cosmetic detail: in the standard logistic predator–prey extension, the coexistence state becomes an attracting equilibrium rather than a neutrally stable center.

Second, the closed orbits imply that populations can recover from arbitrarily small positive densities. This is a deterministic feature of the equations, not a robust ecological fact. In real populations, very low abundance can mean demographic stochasticity, Allee effects, genetic loss, or local extinction.

Third, all prey are identical. There is no defended prey type and no undefended prey type. Therefore the model cannot represent a defense–growth trade-off, prey trait composition, or evolutionary sorting.

Fourth, the system is well mixed. The variables $X(t)$ and $Y(t)$ describe total population densities, not spatial distributions. The model cannot ask whether predators and edible prey overlap in space, or whether diffusion-driven patterns help or harm predator persistence.

Fifth, there is no explicit rescue threshold. The predator either persists in the closed-orbit dynamics or is absent from the model. To study indirect evolutionary rescue, we need a parameter such as predator mortality and a criterion for whether the predator population avoids extinction.

These limitations are not failures of the Lotka–Volterra model. They tell us what must be added. The rest of this report keeps the same modelling discipline, but adds the ingredients required for the research question: prey heterogeneity, predator stress, spatial movement, and a comparison between non-spatial and spatial persistence thresholds.

3 The Extended Lotka–Volterra Model

3.1 From One Prey Type to Two

The Lotka–Volterra system has one prey population and one predator population. Our problem requires one additional biological layer: prey are not all equivalent. We split the prey into two types:

$$U(t) = \text{undefended prey}, \quad D(t) = \text{defended prey}, \quad (3.1)$$

and keep

$$P(t) = \text{predator}. \quad (3.2)$$

The total prey density is

$$N(t) = U(t) + D(t). \quad (3.3)$$

The distinction between U and D is defined by a trade-off. Undefended prey are more edible but grow faster. Defended prey are less edible but pay a growth or competitive cost. We write this as

$$r_U > r_D, \quad a_U > a_D. \quad (3.4)$$

Here r_U and r_D are prey growth rates, while a_U and a_D are predator attack rates on undefended and defended prey respectively. This two-type reduction is a minimal representation of the defence–growth trade-off observed in rotifer–algal systems and related eco-evolutionary experiments [5, 6].

The model will be used in two stages. First, the ODE model defines the well-mixed baseline and the threshold m_c^{ODE} . Second, the PDE model adds diffusion and defines m_c^{PDE} . The spatial model is therefore evaluated relative to the non-spatial baseline, not as an isolated pattern-forming system.

3.2 A Non-Spatial Baseline

The first extension of Lotka–Volterra is to replace unlimited prey growth by logistic growth and to let the predator consume two prey types. A minimal non-spatial model is

$$\boxed{\begin{aligned} \frac{dU}{dt} &= r_U U \left(1 - \frac{U + D}{K}\right) - a_U U P, \\ \frac{dD}{dt} &= r_D D \left(1 - \frac{U + D}{K}\right) - a_D D P, \\ \frac{dP}{dt} &= e(a_U U + a_D D)P - mP. \end{aligned}} \quad (3.5)$$

This is the direct analogue of Lotka–Volterra for our setting. The prey equations contain logistic growth and predation loss. The predator equation says that predators increase by converting consumed prey into predator biomass or offspring, and decrease at mortality rate m .

The parameter m is important because it will become the environmental stress parameter. Increasing m makes predator persistence harder. In a rescue problem, the central question is not merely whether the equations oscillate or equilibrate, but whether the predator avoids extinction when m is high.

3.3 Evolutionary Sorting

Although (3.5) contains no explicit mutation term, it still contains an evolutionary component if U and D represent heritable prey types already present in the population. The evolutionary variable is the prey composition:

$$q_U = \frac{U}{U + D}, \quad q_D = \frac{D}{U + D}. \quad (3.6)$$

In this minimal model, evolution is represented at the level of prey-type frequency dynamics rather than allele-level population genetics. To see how selection acts, compare the per capita growth rates of the two prey types. For positive U and D ,

$$\begin{aligned}\frac{d}{dt} \log \left(\frac{U}{D} \right) &= \frac{1}{U} \frac{dU}{dt} - \frac{1}{D} \frac{dD}{dt} \\ &= (r_U - r_D) \left(1 - \frac{U + D}{K} \right) - (a_U - a_D)P.\end{aligned}\tag{3.7}$$

This equation is the simplest expression of the rescue mechanism. When predator density is high, the second term is large and defended prey are favoured. When predator density falls, selection against undefended prey weakens, and the faster growth of undefended prey can increase the edible component of the prey community. Predator recovery can then occur indirectly through prey compositional change [3].

3.4 A Coexistence Caveat

There is a mathematical point that matters for the rest of the project. In the pure sorting model (3.5), a positive equilibrium with $U^* > 0$, $D^* > 0$, and $P^* > 0$ is not generic. If all three populations are positive at equilibrium, then the first two equations imply

$$r_U \left(1 - \frac{U^* + D^*}{K} \right) = a_U P^*, \quad r_D \left(1 - \frac{U^* + D^*}{K} \right) = a_D P^*.\tag{3.8}$$

Therefore a coexistence equilibrium with both prey types requires

$$\frac{r_U}{a_U} = \frac{r_D}{a_D}.\tag{3.9}$$

This is a fine-tuning condition, not a robust biological assumption. Thus (3.5) is useful for studying transient sorting and rescue, but it is not the safest baseline for a Turing analysis around a homogeneous coexistence state.

For this reason, the working model includes a small conversion term between prey types. This is a phenomenological modelling device: it keeps both prey states present so that a homogeneous coexistence equilibrium can be used for local stability and Turing analysis. Biologically, such terms could be motivated by mutation, reversible phenotypic switching, inducible defence, or unresolved standing variation, but those mechanisms are not equivalent and are not distinguished in this minimal model. Let μ_{UD} be the rate from undefended to defended prey, and μ_{DU} the rate from defended to undefended prey. The non-spatial working model is

$$\boxed{\begin{aligned}\frac{dU}{dt} &= r_U U \left(1 - \frac{U + D}{K} \right) - a_U U P + \mu_{DU} D - \mu_{UD} U, \\ \frac{dD}{dt} &= r_D D \left(1 - \frac{U + D}{K} \right) - a_D D P + \mu_{UD} U - \mu_{DU} D, \\ \frac{dP}{dt} &= e(a_U U + a_D D)P - mP.\end{aligned}}\tag{3.10}$$

When $\mu_{UD} = \mu_{DU} = 0$, this reduces to the pure sorting model. When one prey type is removed and the carrying capacity is sent formally to infinity, the structure reduces to the classical Lotka–Volterra predator–prey feedback.

3.5 Spatial Extension

The research question is spatial: we want to know whether pattern formation changes the predator rescue threshold. Roy et al. provide the closest spatial precursor by studying defended and undefended prey in a reaction–diffusion eco-evolutionary model with Turing analysis [7]. The present model is deliberately simpler: it keeps the defended–undefended prey structure but focuses on comparing predator persistence thresholds between the ODE and PDE systems. We therefore allow each population to move randomly in space. Let $U(x, t)$, $D(x, t)$, and $P(x, t)$ be densities on a habitat $\Omega \subset \mathbb{R}^n$. The reaction–diffusion version of (3.10) is

$$\begin{aligned} \frac{\partial U}{\partial t} &= \delta_U \Delta U + r_U U \left(1 - \frac{U + D}{K} \right) - a_U U P + \mu_{DU} D - \mu_{UD} U, \\ \frac{\partial D}{\partial t} &= \delta_D \Delta D + r_D D \left(1 - \frac{U + D}{K} \right) - a_D D P + \mu_{UD} U - \mu_{DU} D, \\ \frac{\partial P}{\partial t} &= \delta_P \Delta P + e(a_U U + a_D D)P - mP. \end{aligned} \quad (3.11)$$

Here δ_U , δ_D , and δ_P are diffusion coefficients. We use the letter δ for diffusion to avoid confusing diffusion parameters with the defended prey variable D . A closed habitat is represented by no-flux boundary conditions,

$$\nabla U \cdot \mathbf{n} = \nabla D \cdot \mathbf{n} = \nabla P \cdot \mathbf{n} = 0 \quad \text{on } \partial\Omega. \quad (3.12)$$

The model (3.11) is the extended Lotka–Volterra model for this study. It keeps the predator–prey feedback of the classical equations, but adds the four ingredients needed to test the proposed pattern-mediated rescue mechanism: logistic resource limitation, defended and undefended prey, predator mortality stress, and spatial movement.

3.6 Predator Rescue Threshold

The predator per capita growth rate in the non-spatial model is

$$\frac{1}{P} \frac{dP}{dt} = e(a_U U + a_D D) - m. \quad (3.13)$$

Thus predator persistence depends not only on total prey density, but also on prey composition. For fixed total prey density, a community dominated by undefended prey supports higher predator growth than a community dominated by defended prey.

This motivates the central comparison of the project. Let m_c^{ODE} be the largest predator mortality for which the non-spatial system maintains predators above an extinction threshold, and let m_c^{PDE} be the corresponding threshold in the spatial system. The proposed pattern-mediated effect is measured by

$$\Delta m_c = m_c^{\text{PDE}} - m_c^{\text{ODE}}. \quad (3.14)$$

If $\Delta m_c > 0$, spatial patterning promotes indirect rescue. If $\Delta m_c < 0$, spatial patterning inhibits rescue.

3.7 Operational Definition of Persistence

In deterministic ODE and PDE models, extinction is not always represented by an exact finite-time crossing to zero. For numerical work, predator persistence must therefore be defined operationally. For a final time T and a small extinction threshold $\varepsilon > 0$, predators are considered persistent in the ODE model if

$$P(T) > \varepsilon. \quad (3.15)$$

In the PDE model, predators are considered persistent if the total predator biomass satisfies

$$\int_{\Omega} P(x, T) dx > \varepsilon. \quad (3.16)$$

The thresholds m_c^{ODE} and m_c^{PDE} should be computed using this criterion, and sensitivity to T and ε should be checked.

4 Turing Pattern Formation

4.1 Why Diffusion Can Create Patterns

Diffusion is usually thought of as a smoothing process. If one population is high in one place and low in another, random movement tends to reduce the difference. This makes Turing pattern formation initially surprising: under suitable conditions, diffusion can destabilize a homogeneous state that would otherwise be stable [1, 8]. The outcome is not uniform spreading, but the emergence of a spatial pattern with a characteristic wavelength.

The key point is that diffusion acts on a system with several interacting variables. A single diffusing population with stable local dynamics will not generate a Turing pattern. But if one component locally amplifies a perturbation while another suppresses it, and if the two components move at sufficiently different rates, diffusion can turn a stable local balance into a spatially patterned state.

4.2 The Two-Variable Calculation

The cleanest derivation uses two variables. Consider

$$\frac{\partial u}{\partial t} = f(u, v) + D_1 \Delta u, \quad (4.1)$$

$$\frac{\partial v}{\partial t} = g(u, v) + D_2 \Delta v. \quad (4.2)$$

Suppose the non-spatial system has a homogeneous equilibrium (u^*, v^*) satisfying

$$f(u^*, v^*) = 0, \quad g(u^*, v^*) = 0. \quad (4.3)$$

Let

$$J = \begin{pmatrix} f_u & f_v \\ g_u & g_v \end{pmatrix}_{(u^*, v^*)} \quad (4.4)$$

be the Jacobian of the reaction terms at the equilibrium. The equilibrium is stable in the well-mixed ODE if

$$\text{tr } J = f_u + g_v < 0, \quad \det J = f_u g_v - f_v g_u > 0. \quad (4.5)$$

Now perturb the equilibrium by a spatial Fourier mode:

$$u(x, t) = u^* + \hat{u}(t)e^{ik \cdot x}, \quad v(x, t) = v^* + \hat{v}(t)e^{ik \cdot x}. \quad (4.6)$$

After linearization, this mode evolves according to the modified Jacobian

$$J_k = \begin{pmatrix} f_u - D_1 k^2 & f_v \\ g_u & g_v - D_2 k^2 \end{pmatrix}. \quad (4.7)$$

The trace becomes more negative:

$$\text{tr } J_k = \text{tr } J - (D_1 + D_2)k^2. \quad (4.8)$$

The determinant, however, is

$$\det J_k = \det J - k^2(D_1 g_v + D_2 f_u) + D_1 D_2 k^4. \quad (4.9)$$

Thus diffusion can destabilize the equilibrium if $\det J_k < 0$ for some nonzero wavenumber k . Equivalently, the quadratic in k^2 must dip below zero. A standard form of the Turing condition is

$$D_1 g_v + D_2 f_u > 0, \quad (D_1 g_v + D_2 f_u)^2 > 4D_1 D_2 \det J. \quad (4.10)$$

These inequalities say that the equilibrium is stable to homogeneous perturbations but unstable to some spatially heterogeneous perturbations.

4.3 Different Diffusion Rates

The instability requires sufficiently different diffusion rates. If $D_1 = D_2 = D$, then

$$J_k = J - Dk^2 I. \quad (4.11)$$

The eigenvalues are simply shifted by $-Dk^2$, so diffusion only makes the stable equilibrium more stable. Turing instability needs differential movement. In the standard activator–inhibitor interpretation, one component locally amplifies a perturbation, while another component suppresses it and diffuses faster. Local activation combined with long-range inhibition can produce spatial structure.

4.4 Pattern Wavelength and Boundaries

The linear calculation identifies which spatial modes grow. The fastest-growing mode has wavenumber $k_{\max}$ and gives an expected pattern length scale

$$\ell_* \approx \frac{2\pi}{k_{\max}}. \quad (4.12)$$

This does not by itself determine the final pattern. Spots, stripes, labyrinths, or patchy structures depend on the nonlinear dynamics after the instability grows.

On a finite domain with no-flux boundary conditions, the allowed values of k^2 are not continuous. They are eigenvalues of the Laplacian on the habitat. A pattern appears only if one or more allowed modes lie in the unstable window. This matters for ecology: habitat size and shape can determine whether a spatial instability is visible, and whether the resulting pattern resembles stripes, spots, or isolated patches.

4.5 How This Enters the Rescue Problem

The two-variable calculation is useful because it shows the mechanism clearly. Our working model has three interacting variables, so the Turing conditions no longer reduce to the simple trace–determinant inequalities above. In practice, the stability test is spectral: for each admissible spatial mode, inspect the eigenvalues of the diffusion-modified Jacobian.

For our model, define

$$\mathbf{w} = (U, D, P)^T \quad (4.13)$$

and let $\mathbf{F}(\mathbf{w})$ be the reaction part of (3.11). At a homogeneous equilibrium $\mathbf{w}^*$, the Jacobian of the reaction terms is

$$J = \left. \frac{\partial \mathbf{F}}{\partial \mathbf{w}} \right|_{\mathbf{w}^*}. \quad (4.14)$$

A spatial perturbation of wavenumber k satisfies

$$\lambda \mathbf{q} = [J - k^2 \text{diag}(\delta_U, \delta_D, \delta_P)] \mathbf{q}, \quad (4.15)$$

or equivalently

$$\det(J - k^2 \text{diag}(\delta_U, \delta_D, \delta_P) - \lambda I) = 0. \quad (4.16)$$

A Turing instability occurs when all eigenvalues of J have negative real parts, but there exists some $k > 0$ for which $J - k^2 \text{diag}(\delta_U, \delta_D, \delta_P)$ has an eigenvalue with positive real part.

This is the mathematical bridge to the biological question. Spatial patterning may create local patches where predators and undefended prey overlap, raising predator growth. It may also separate predators from edible prey, lowering predator growth. Turing analysis is therefore not the final result. It identifies whether spatial patterns can arise; the rescue question is answered only after comparing m_c^{PDE} with m_c^{ODE} .

5 Planned Analysis

1. **Non-spatial baseline.** Set $\delta_U = \delta_D = \delta_P = 0$. Find homogeneous equilibria and determine the predator persistence threshold m_c^{ODE} .
2. **Local stability.** Linearize the reaction terms around a positive homogeneous equilibrium and compute the Jacobian J .
3. **Turing instability.** Add diffusion and scan the eigenvalues of

$$J - k^2 \text{diag}(\delta_U, \delta_D, \delta_P) \quad (5.1)$$

over admissible spatial modes k . Identify parameter regions where the ODE equilibrium is stable but spatial modes are unstable.

4. **Spatial threshold.** Simulate the PDE model with no-flux boundary conditions and compute m_c^{PDE} .
5. **Threshold comparison.** Evaluate

$$\Delta m_c = m_c^{\text{PDE}} - m_c^{\text{ODE}}. \quad (5.2)$$

Use this to classify whether spatial patterning promotes rescue, inhibits rescue, or has a parameter-dependent effect.

6. **Mechanistic diagnostics.** Measure total predator biomass, spatial overlap between predators and undefended prey, and mean edible-prey availability

$$\langle a_U U + a_D D \rangle. \quad (5.3)$$

These diagnostics should explain why Δm_c is positive or negative.

7. **Robustness checks.** Vary initial prey composition, extinction threshold, final time T , domain size, diffusion ratios, conversion rates, and at least one saturating functional response such as Holling type II predation.

6 Expected Contributions and Limitations

The expected contribution is a focused threshold comparison. Rather than claiming that pattern-mediated indirect evolutionary rescue is already established, the study tests whether a spatially patterned predator–prey system can support predators at mortality levels that would cause loss in the corresponding non-spatial model. The main scientific outcome is not the existence of a spatial pattern itself, but whether that pattern changes the predator persistence threshold relative to the well-mixed system.

There are three possible outcomes. If $\Delta m_c > 0$, spatial patterning expands the predator persistence window and acts as a rescue-promoting mechanism. If $\Delta m_c < 0$, patterning separates predators from edible prey or otherwise lowers effective predator

growth, so spatial structure inhibits rescue. If Δm_c changes sign across parameter regimes, the main result is conditional: pattern formation can either help or harm predator persistence depending on diffusion rates, prey defence costs, and spatial overlap.

The main limitation is that the conversion terms between U and D are phenomenological. They make coexistence and Turing analysis mathematically well posed, but they do not by themselves identify whether the biological source is mutation, plastic switching, inducible defence, or persistent standing variation. A later model could separate these mechanisms once the minimal threshold question is understood. Similarly, the baseline model uses mass-action predation; robustness checks with a saturating functional response are needed before treating the threshold shift as a general ecological prediction.

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